Distributions and Maximum Likelihood

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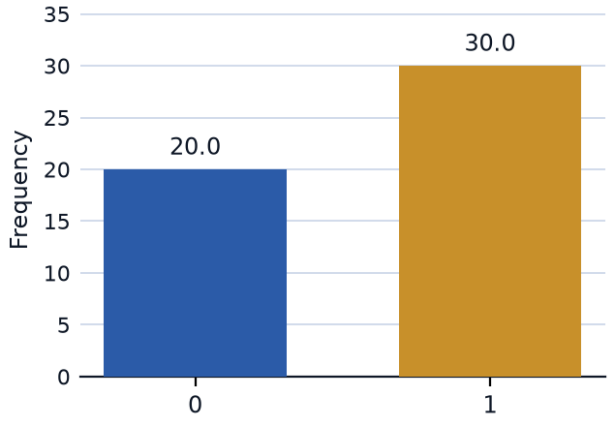
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$? Show your complete algebraic work.

2 [Foundational] • Target: Discrete Random Variables and Bernoulli Likelihood

A dataset consists of observations $x_1, x_2, \ldots, x_n$ that are assumed to be drawn independently from a Bernoulli distribution with parameter p. The probability mass function is given by $P(X = k; p) = p^k(1 - p)^{1-k}$ for $k \in \{0, 1\}$. The following bar chart shows the frequency distribution of these observations, where $n = 50$. { "type": "bar", "labels": ["0", "1"], "data": [20, 30], "title": "Frequency Distribution of Bernoulli Observations" } Using the method of Maximum Likelihood Estimation (MLE), determine the estimated value of p for this distribution.



3 [Foundational] • Target: Continuous Gaussian Normal Distributions

Consider a sequence of n independent Bernoulli trials, where the probability of success in each trial is p. Let X be the random variable representing the number of successes in these n trials. Thus, $X \sim \text{Binomial}(n, p)$. We are given a dataset of N observed values $\{x_1, x_2, \ldots, x_N\}$ for X, where each x_i is an observed count of successes from n trials. The observed data are: [3, 4, 2, 5, 4, 3, 2, 4, 5, 3]. If we assume that $n = 5$ (the number of trials for each observation is fixed at 5), find the value of p that maximizes the likelihood of observing this dataset. Then, compute the maximum likelihood estimate (MLE) of the probability of success, p, and round your answer to two decimal places.

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4 [Foundational] • Target: The Likelihood Function and Log-Likelihood Formulation

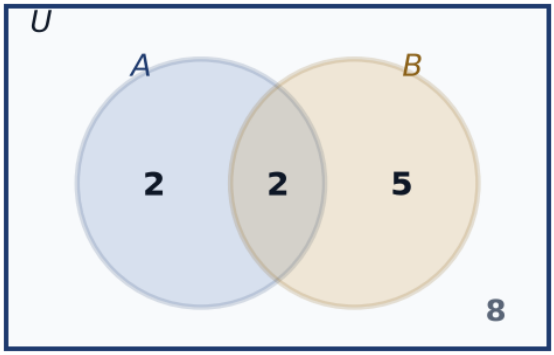
A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$. Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ. Part 2: Set the derivative to zero to find the MLE of λ.

5 [Foundational] • Target: Properties of Estimators: Bias and Consistency

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2? Show your complete algebraic work.

6 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Consider a dataset of observations $x_1, x_2, \ldots, x_n$ assumed to be drawn independently from an exponential distribution with probability density function (PDF) $f(x; \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$ and $\lambda > 0$. The likelihood function for this dataset is given by $L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$. To find the Maximum Likelihood Estimator (MLE) for λ, we first consider the log-likelihood function $\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$. If the observed data points are $\{1.2, 2.5, 0.8, 3.1, 1.5 \backslash)$, and $n = 5$, determine the MLE of λ.



7 [Foundational] • Target: Discrete Random Variables and Bernoulli Likelihood

A dataset consists of $n = 100$ independent and identically distributed (i.i.d.) observations $x_1, x_2, \ldots, x_{100}$. These observations are drawn from a Bernoulli distribution with parameter p. The probability mass function (PMF) of a Bernoulli distribution is given by $P(X = x; p) = p^x (1 - p)^{1 - x}$ for $x \in \{0, 1\}$. The likelihood function $L(p)$ for this dataset is the product of the PMFs of the individual observations. If we observe $k = 60$ successes (i.e., $x_i = 1$) and $n - k = 40$ failures (i.e., $x_i = 0$), what is the likelihood function $L(p)$ for p? Show your complete algebraic work.

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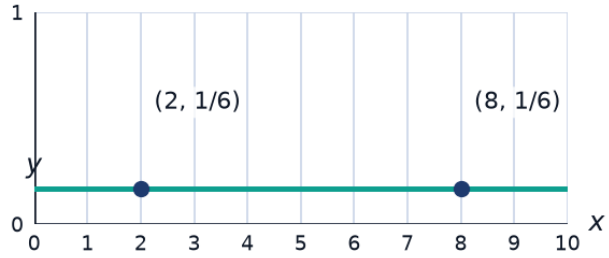
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Continuous Gaussian Normal Distributions**

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k? Show your complete algebraic work.

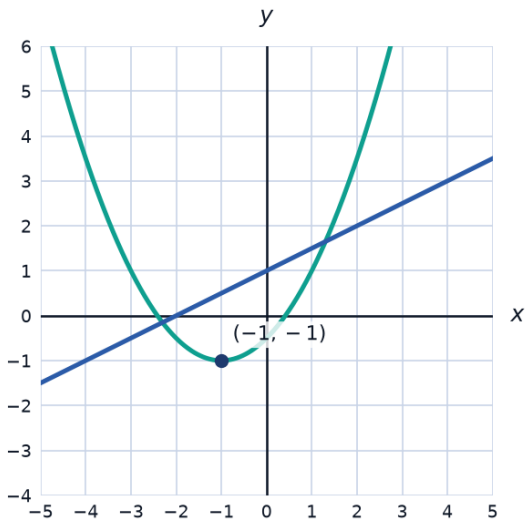
9 **[Intermediate]** • **Target: The Likelihood Function and Log-Likelihood Formulation**

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$? Show your complete algebraic work.



10 **[Intermediate]** • **Target: Properties of Estimators: Bias and Consistency**

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$? Show your complete algebraic work.



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11 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let X be the first integer generated and Y be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

12 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)? Show your complete algebraic work.

13 [Intermediate] • Target: Continuous Gaussian Normal Distributions

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario? Show your complete algebraic work.

14 [Intermediate] • Target: The Likelihood Function and Log-Likelihood Formulation

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

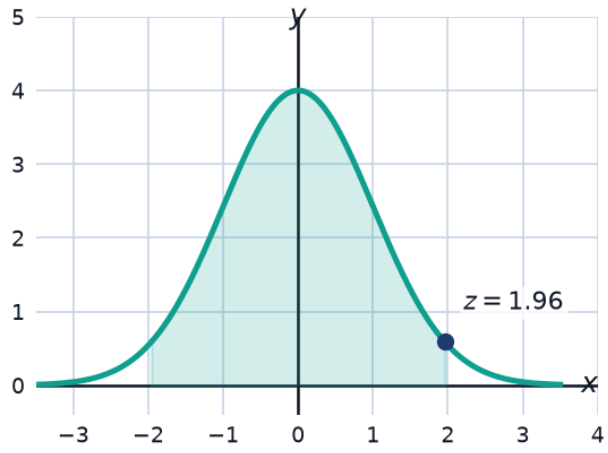
15 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: [illegible] Linear Regression and OLS

Given $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

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16 **[Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

Given $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.



17 **[Challenging] • Target: Discrete Random Variables and Bernoulli Likelihood • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Given $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

18 **[Challenging] • Target: Continuous Gaussian Normal Distributions • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

19 **[Challenging] • Target: The Likelihood Function and Log-Likelihood Formulation • Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Given $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

20 **[Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 3.01 Gradient Descent and Optimization**

Expected value E(X) for discrete X:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(5, 4)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 5, \; \sigma^2 = 4$

b. $\mu = 4, \; \sigma^2 = 5$

c. $\mu = 5, \; \sigma^2 = 2$

d. $\mu = 2, \; \sigma^2 = 4$

22 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(10, 9)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 10, \; \sigma^2 = 3$

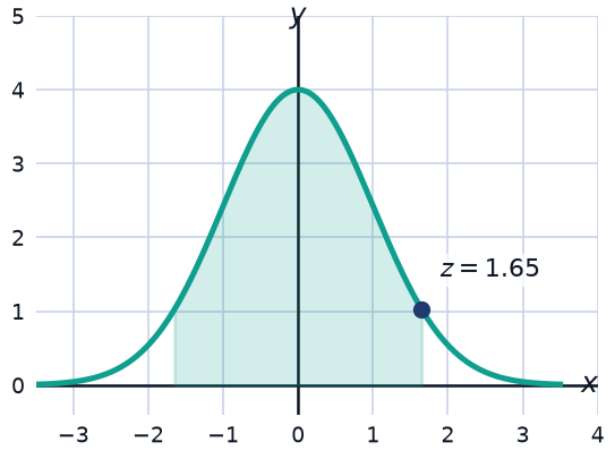
b. $\mu = 9, \; \sigma^2 = 10$

c. $\mu = 10, \; \sigma^2 = 9$

d. $\mu = 3, \; \sigma^2 = 9$

23 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(15, 16)$. What is the mean μ and variance σ^2 of this feature?



a. $\mu = 15, \; \sigma^2 = 16$

b. $\mu = 16, \; \sigma^2 = 15$

c. $\mu = 15, \; \sigma^2 = 4$

d. $\mu = 4, \; \sigma^2 = 16$

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24 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(20, 25)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 20, \; \sigma^2 = 25$

b. $\mu = 25, \; \sigma^2 = 20$

c. $\mu = 20, \; \sigma^2 = 5$

d. $\mu = 5, \; \sigma^2 = 25$

25 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 25, \; \sigma^2 = 36$

b. $\mu = 36, \; \sigma^2 = 25$

c. $\mu = 25, \; \sigma^2 = 6$

d. $\mu = 6, \; \sigma^2 = 36$

26 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(30, 49)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 30, \; \sigma^2 = 49$

b. $\mu = 49, \; \sigma^2 = 30$

c. $\mu = 30, \; \sigma^2 = 7$

d. $\mu = 7, \; \sigma^2 = 49$

27 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

a. $\hat{\mu}_{\text{MLE}} = 3$

b. $\hat{\mu}_{\text{MLE}} = 6$

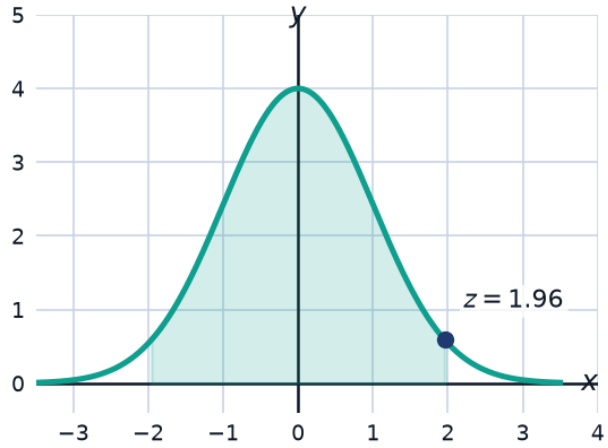
c. $\hat{\mu}_{\text{MLE}} = 12$

d. $\hat{\mu}_{\text{MLE}} = 1.5$

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28 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Given \(n = 8 \) independent observations from a normal distribution with sum \(\sum_{i=1}^{8} x_i = 24 \), what is the Maximum Likelihood Estimate (MLE) \(\hat{\mu}_{\text{MLE}} \)?



a. \(\hat{\mu}_{\text{MLE}} = 3 \)

b. \(\hat{\mu}_{\text{MLE}} = 6 \)

c. \(\hat{\mu}_{\text{MLE}} = 24 \)

d. \(\hat{\mu}_{\text{MLE}} = 1.5 \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

In a binary click-through model, an ad is clicked $k = 40$ times out of $n = 200$ impressions. What is the Maximum Likelihood Estimate $\hat{p}_{MLE}$ of the click probability?

a. 0.20

b. 0.25

c. 0.16

d. 0.40

30 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

What is the variance of a Bernoulli random variable with success probability $p = 0.3$?

a. 0.21

b. 0.30

c. 0.70

d. 0.49

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31 **[Intermediate]** • **Target: Discrete Random Variables and Bernoulli Likelihood**

Show how maximizing the Bernoulli log-likelihood relates to the binary cross-entropy loss function used in neural networks.

a. Maximizing Bernoulli log-likelihood is mathematically identical to minimizing binary cross-entropy

b. Cross-entropy is the reciprocal of the log-likelihood

c. They only match when $p = 0.5$

d. Cross-entropy ignores negative classes

32 **[Intermediate]** • **Target: Continuous Gaussian Normal Distributions**

A feature vector component follows $X \sim \mathcal{N}(50, 16)$. What are its mean μ and standard deviation σ?

a. $\mu = 50, \quad \sigma = 4$

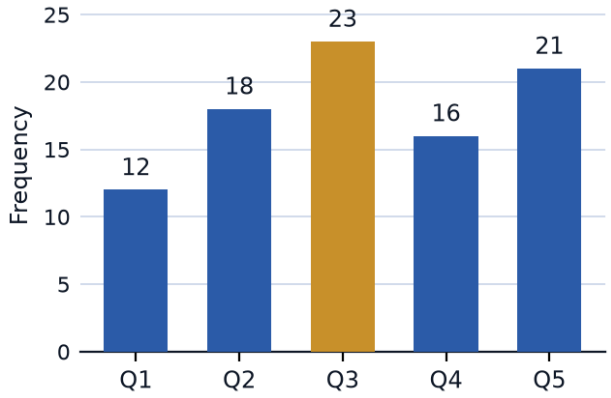
b. $\mu = 50, \quad \sigma = 16$

c. $\mu = 16, \quad \sigma = 50$

d. $\mu = 25, \quad \sigma = 4$

33 **[Intermediate]** • **Target: Continuous Gaussian Normal Distributions**

By the empirical $(68 - 95 - 99.7)$ rule, approximately what percentage of samples from $\mathcal{N}(\mu, \sigma^2)$ fall within the interval $(\mu - 2\sigma, \mu + 2\sigma)$?



a. 95.4%

b. 68.2%

c. 99.7%

d. 50.0%

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34 **[Intermediate]** • **Target: Continuous Gaussian Normal Distributions**

Standardize the value $x = 62$ from the distribution $\mathcal{N}(50, 16)$ into a standard normal z-score.

a. $z = 3.0$

b. $z = 0.75$

c. $z = 12$

d. $z = 1.5$

35 **[Intermediate]** • **Target: The Likelihood Function and Log-Likelihood Formulation**

Why is the log-likelihood $\ln L(\theta) = \sum_{i=1}^{n} \ln p(x_i|\theta)$ preferred over raw likelihood in machine learning optimization?

a. It transforms products of probabilities into sums, preventing numerical underflow and simplifying derivatives

b. It converts non-convex problems into concave problems unconditionally

c. It eliminates the need for normalization

d. It ensures the sample mean is zero

36 **[Intermediate]** • **Target: The Likelihood Function and Log-Likelihood Formulation**

For n independent samples from $\mathcal{N}(\mu, \sigma^2)$, what is the analytical expression for the log-likelihood?

a. $-\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$

b. $-\frac{n}{2}\ln(2\pi) - \sum_{i=1}^{n}(x_i - \mu)$

c. $\frac{n}{2\sigma^2}\sum(x_i - \mu)^2$

d. $-\frac{1}{\sigma^2}\sum(x_i - \mu)$

37 **[Intermediate]** • **Target: The Likelihood Function and Log-Likelihood Formulation**

Given 5 observations $\{2, 4, 4, 6, 9\}$, compute the sample mean $\bar{x}$, which is the Gaussian MLE $\hat{\mu}_{MLE}$.

a. 5.0

b. 4.0

c. 4.6

d. 5.5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Why is the Maximum Likelihood Estimator for variance $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \bar{x})^2$ biased for finite samples?

a. Its expected value is $\frac{n-1}{n}\sigma^2 < \sigma^2$, underestimating true variance due to using sample mean $\bar{x}$

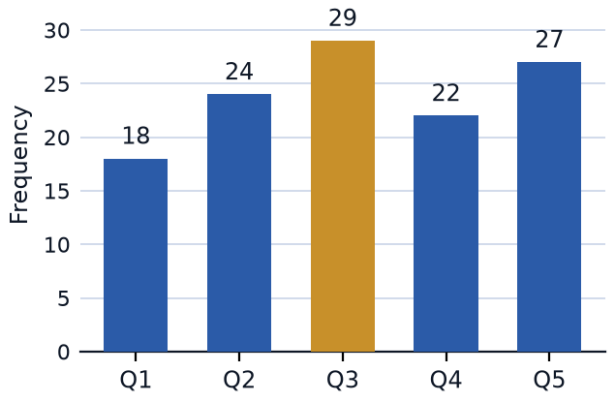
b. It only works for standard normal data

c. Its expected value approaches infinity

d. It ignores outliers

39 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

What is Bessel's correction to make the sample variance estimator unbiased?



a. Multiply by $\frac{n}{n-1}$

b. Multiply by $\frac{n-1}{n}$

c. Add $\frac{1}{n}$

d. Subtract the mean

40 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 3.01 Gradient Descent and Optimization

What does it mean for an estimator $\hat{\theta}_n$ to be asymptotically consistent?

a. As sample size $n \to \infty$, $\hat{\theta}_n$ converges in probability to the true parameter θ

b. The estimator has zero variance for all sample sizes

c. The estimator is unbiased for $n = 1$

d. The estimator follows a student-t distribution

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41 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 25,\ \sigma^2 = 36$

b. $\mu = 36,\ \sigma^2 = 25$

c. $\mu = 25,\ \sigma^2 = 6$

d. $\mu = 6,\ \sigma^2 = 36$

42 [Challenging] • Target: Discrete Random Variables and Bernoulli Likelihood • Cross-Lesson: 6.01 Linear Regression and OLS

A Normal random variable has $\mu = 170$ and $\sigma = 15$. Find $P(148 < X < 190)$.

a. 0.5376

b. 0.8376

c. 0.7376

d. 0.6376

43 [Challenging] • Target: Continuous Gaussian Normal Distributions • Cross-Lesson: 7.01 Logistic Regression and Classification

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. n is large (n≥30)

c. σ is known

d. p<0.5

44 [Challenging] • Target: The Likelihood Function and Log-Likelihood Formulation • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A Normal random variable has $\mu = 111$ and $\sigma = 10$. Find $P(102 < X < 116)$.

a. 0.5074

b. 0.3074

c. 0.2074

d. 0.4074

45 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A Normal random variable has $\mu = 114$ and $\sigma = 10$. Find $P(100 < X < 119)$.

a. 0.6107

b. 0.3107

c. 0.5107

d. 0.4107

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